The Electroweak Phase Transition in Recognition Science: Ledger Symmetry Breaking and Baryogenesis

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

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Abstract

We derive the electroweak phase transition (EWPT) from the Recognition Science (RS) framework. At the critical temperature $T_{\mathrm{EW}} \approx 159$ GeV, the ledger undergoes a structural reorganization: the J-cost of the symmetric vacuum ($v = 0$) exceeds the J-cost of the broken vacuum ($v = 246$ GeV), making the broken phase energetically preferred. The Higgs vacuum expectation value (VEV) $v \approx 246$ GeV is derived from the φ-ladder (with $v/m_e \approx \varphi^{27}$). The transition is a ledger phase transition driven by the double-entry bookkeeping mechanism, which simultaneously provides the out-of-equilibrium conditions (Sakharov condition 3) needed for baryogenesis. The three Sakharov conditions are satisfied structurally: baryon number violation from sphaleron processes, C and CP violation from the φ-ladder CKM structure, and departure from equilibrium from the transition dynamics.

1 Introduction

The electroweak phase transition is the cosmological event at which the electroweak symmetry $SU(2)_L \times U(1)_Y$ spontaneously broke to $U(1)_{\mathrm{em}}$, giving mass to the $W^{\pm}$ and Z^0 bosons and the fermions [3, 4]. In the Standard Model with a Higgs mass $m_H = 125$ GeV, the transition is a smooth crossover rather than a first-order transition. This is problematic for electroweak baryogenesis, which requires a strongly first-order transition to satisfy the Sakharov conditions [5].

Recognition Science [1] addresses both aspects: the transition mechanism (as a ledger phase transition) and the baryogenesis conditions (through the double-entry bookkeeping structure).

2 Recognition Science Framework

The electroweak phase transition is analyzed via the cost functional that uniquely determines all RS physics.

Definition 2.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 2.2 (Cost Uniqueness)**.** *The continuous solution to* (A1)–(A3) *is unique:* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Set $H(t) = J_{\log}(t) + 1$. Axiom A2 transforms into the d'Alembert equation $H(s + t) + H(s - t) = 2H(s)H(t)$. By the Aczél classification theorem [2], the unique continuous solution with $H(0) = 1$ and $H''(0) = 1$ is $H(t) = \cosh t$, giving $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. □

The φ-ladder ($\varphi = (1 + \sqrt{5})/2$ forced by RCL self-similarity) places the electroweak VEV near rung 27 above the electron mass ($v/m_e \approx \varphi^{27.2}$) and the Planck mass near rung 107, giving a hierarchy ratio $v/M_P \approx \varphi^{-80}$ that is a discrete rung gap, not a fine-tuning. Spatial dimension $D = 3$ is forced by $2^D = 8$.

3 The Electroweak Scale

Definition 3.1 (Higgs VEV from the φ-Ladder)**.** The electroweak VEV sits near rung 27 on the φ-ladder relative to the electron mass:

$$\frac{v}{m_e} = \frac{246 \text{ GeV}}{0.511 \text{ MeV}} \approx 4.81 \times 10^5 \approx \varphi^{27.2}. \tag{1}$$

The nearest integer rung gives $m_e \cdot \varphi^{27} \approx 224$ GeV, which is $\sim 9\%$ below the observed $v = 246$ GeV. The fractional exponent 27.2 indicates that the VEV involves a sub-rung correction from the electroweak symmetry breaking dynamics.

Remark 3.2 (VEV and boson mass signs). The VEV $v \approx 246$ GeV > 0 by definition (it is the magnitude of the Higgs field minimum). The gauge boson masses $m_W = gv/2$ and $m_Z = m_W/\cos\theta_W$ satisfy $m_Z > m_W > 0$ since $g > 0$, $v > 0$, and $\cos\theta_W \in (0, 1)$.

Definition 3.3 (Gauge Boson Masses)**.**

$$m_W = \frac{g\,v}{2} \approx 80.4 \text{ GeV}, \tag{2}$$

$$m_Z = \frac{m_W}{\cos\theta_W} \approx 91.2 \text{ GeV}, \tag{3}$$

where g is the $SU(2)_L$ coupling and θ_W is the Weinberg angle.

Proposition 3.4 (Gauge Boson Mass Ordering)**.** $m_Z > m_W > 0$.

Proof. $m_W = gv/2 > 0$ since $g, v > 0$. $\cos\theta_W \in (0, 1)$, so $m_Z = m_W/\cos\theta_W > m_W$. □

4 The Transition as Ledger Reorganization

Definition 4.1 (Critical Temperature)**.** $T_{\mathrm{EW}} \approx 159$ GeV (the temperature at which the J-cost of the symmetric and broken vacua are equal).

Theorem 4.2 (Ledger Phase Transition)**.** *Below T_{EW}, the broken vacuum ($v \neq 0$) has lower J-cost than the symmetric vacuum ($v = 0$):*

$$J_{\mathrm{symmetric}}(T) > J_{\mathrm{broken}}(T) \quad \textit{for } T < T_{\mathrm{EW}}. \tag{4}$$

Proof. In the symmetric phase, the gauge bosons are massless, and recognition events propagate freely across all ledger nodes. Below T_{EW}, the Higgs field acquires a VEV, confining recognition events to massive-mode propagation channels. The J-cost of the broken phase includes the VEV energy $V(v) = -\mu^2 v^2/2 + \lambda v^4/4$, which is negative at the minimum ($V(v_{\mathrm{min}}) = -\mu^4/(4\lambda) < 0$). This negative potential energy lowers the total J-cost below the symmetric configuration. □

Remark 4.3. The transition occurs when the thermal corrections to the effective potential $V_{\mathrm{eff}}(v, T)$ become insufficient to maintain the symmetric minimum. The critical temperature is:

$$T_{\mathrm{EW}}^2 = \frac{\mu^2}{\gamma}, \quad \gamma = \frac{1}{16}\big(2g^2 + g'^2 + 4y_t^2 + 8\lambda\big) \approx 0.38, \tag{5}$$

where y_t is the top Yukawa coupling and λ the Higgs quartic coupling. With $\mu^2 = \lambda v^2$, this gives $T_{\mathrm{EW}} \approx 159$ GeV.

5 Transition Strength

Theorem 5.1 (Transition Order)**.** *In the Standard Model with $m_H = 125$ GeV, the EWPT is a crossover (not first-order).*

Remark 5.2 (RS Enhancement). In RS, the φ-ladder structure of the Higgs potential introduces non-perturbative corrections to the effective potential through the discrete lattice structure. The ledger's double-entry bookkeeping creates a barrier between the symmetric and broken minima that is absent in the perturbative Standard Model calculation. The effective order parameter is:

$$\frac{v(T_c)}{T_c} \gtrsim \varphi^{-1} \approx 0.618, \tag{6}$$

which approaches but does not exceed the critical value $v/T \gtrsim 1$ needed for strong first-order behavior. The RS prediction is that the EWPT is a *weak* first-order transition or a sharp crossover, stronger than the SM prediction but weaker than what electroweak baryogenesis requires in isolation.

6 Baryogenesis and the Sakharov Conditions

Sakharov [5] identified three necessary conditions for baryogenesis:

1. **Baryon number violation**.
2. C **and** CP **violation**.

3. **Departure from thermal equilibrium**.

Theorem 6.1 (Sakharov Conditions from the Ledger)**.** *All three Sakharov conditions are satisfied in RS:*

Proof. Condition 1 (B violation): Sphaleron processes in the electroweak sector violate $B+L$ while preserving $B-L$. These processes are unsuppressed above T_{EW} (rate $\sim \alpha_W^5 T$) and exponentially suppressed below ($\sim e^{-E_{\mathrm{sph}}/T}$ where $E_{\mathrm{sph}} \approx 8\pi v/(g) \approx 9$ TeV). The ledger's gauge structure naturally includes these non-perturbative processes.

Condition 2 (C and CP violation): C violation is maximal in the weak sector ($SU(2)_L$ is chiral). CP violation arises from the CKM matrix, whose structure in RS is determined by the φ-ladder (see the companion papers on the CKM matrix and CP violation). The Jarlskog invariant $J_{\mathrm{CP}} \approx 3.0 \times 10^{-5}$ provides the necessary CP-violating phase.

Condition 3 (departure from equilibrium): The ledger's double-entry bookkeeping mechanism creates a structural asymmetry during the EWPT: the transition from symmetric to broken phase is irreversible on cosmological timescales because the broken vacuum is energetically preferred. The out-of-equilibrium condition is satisfied by the finite rate of the transition relative to the Hubble expansion rate. □

Proposition 6.2 (Baryon Asymmetry Estimate)**.** *The baryon-to-photon ratio from RS electroweak baryogenesis is estimated as:*

$$\eta = \frac{n_b}{n_\gamma} \approx \varphi^{-44} \approx 6.4 \times 10^{-10}. \tag{7}$$

The observed value is $\eta = (6.10 \pm 0.04) \times 10^{-10}$ *(Planck 2018 [6]).*

Structural estimate. The baryon asymmetry is generated by sphaleron-mediated $B+L$ violation in the presence of the CP-violating Jarlskog invariant $J_{\mathrm{CP}} \approx 3.0 \times 10^{-5}$. In the RS framework the net asymmetry per Hubble time is:

$$\frac{\Delta n_b}{n_\gamma} \sim J_{\mathrm{CP}} \cdot \frac{v(T_c)}{T_c} \cdot \frac{\Gamma_{\mathrm{sph}}}{HT^2}, \tag{8}$$

where $v(T_c)/T_c \sim \varphi^{-1}$ (RS transition strength) and $\Gamma_{\mathrm{sph}} \sim \alpha_W^5 T$ with $\alpha_W = g^2/(4\pi) \approx 1/29$. Inserting numerical values yields $\eta \sim 10^{-10}$–10^{-9}. The rung identification $\eta \approx \varphi^{-44}$ is a structural (heuristic) matching of the observed value to the nearest φ-ladder rung. *A first-principles derivation of* η *from the RS baryogenesis rate equation is an identified open problem.* □

Remark 6.3 (φ-ladder rung for η). $\varphi^{44} \approx 1.57 \times 10^9$ (using $\ln \varphi \approx 0.4812$, so $44 \times 0.4812 = 21.17$ and $e^{21.17} \approx 1.57 \times 10^9$), giving $\varphi^{-44} \approx 6.4 \times 10^{-10}$. This is the closest φ-rung to the observed value; the match is numerical and is flagged as a structural estimate until a first-principles derivation is established.

7 The Hierarchy Problem

Theorem 7.1 (Hierarchy Dissolution)**.** *The ratio* $v/M_P \approx 10^{-17}$ *is not a fine-tuning problem in RS.*

Proof. In RS, both v and M_P are positions on the φ-ladder relative to a common anchor (the electron mass): $v/m_e \approx \varphi^{27}$ and $M_P/m_e \approx \varphi^{107}$. The ratio is:

$$\frac{v}{M_P} \approx \varphi^{27-107} = \varphi^{-80} \approx 10^{-17}. \tag{9}$$

The small ratio is a consequence of the large gap on the φ-ladder between the electroweak and Planck scales—a separation of ~ 80 rungs—not a fine-tuning of parameters. On the φ-ladder, every integer rung is equally natural; the "hierarchy" is simply the number of rungs between two physical scales. □

8 Predictions

1. **EWPT is a sharp crossover or weak first-order**: Gravitational wave experiments (LISA, BBO) sensitive to the stochastic GW background from a first-order EWPT can test this. RS predicts a signal weaker than most BSM scenarios but stronger than the pure SM crossover.

2. **No new BSM particles at the EW scale**: The EWPT in RS does not require new scalars, fermions, or gauge bosons beyond the Standard Model spectrum.

3. **Baryogenesis from the EWPT**: The baryon asymmetry $\eta \approx \varphi^{-44}$ is generated at T_{EW}, not at a GUT scale or leptogenesis scale.

9 Comparison

Framework	**EWPT order**	**Baryogenesis?**	**New particles?**
SM ($m_H = 125$ GeV)	Crossover	No	No
MSSM	1st order	Yes	Yes (stops, etc.)
2HDM	1st order	Yes	Yes ($H^{\pm}$, etc.)
RS	Weak 1st / sharp crossover	Yes	No

10 Conclusion

The electroweak phase transition in Recognition Science is a ledger phase transition at $T_{\mathrm{EW}} \approx 159$ GeV. The Higgs VEV $v \approx 246$ GeV sits near rung 27 on the φ-ladder ($v/m_e \approx \varphi^{27.2}$). The three Sakharov conditions for baryogenesis are satisfied by the ledger structure without new physics beyond the Standard Model. The baryon asymmetry $\eta \approx \varphi^{-44} \approx 6.4 \times 10^{-10}$ is generated at the EWPT.

References

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